WORKED SOLUTION

STEP II 2022, Question 1

Pure • Integration by Parts

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Read the question before you go on. It is in the STEP II paper in OCR's download of the 2022 STEP papers (a zip file):

www.ocr.org.uk/Images/696587-2022-past-papers.zip

The key idea

Every part rests on one move: integrating by parts produces a second integral that cancels, or nearly cancels, something you already have. In (i) and (ii) that means spotting that the integrand is the derivative of a product. In (iii) the same move shows that the difference between the two integrals is just the change in e^x/x between the limits, so the sketch in (a) does the rest of the work.

Solution

Part (i)

Integrate the first term by parts, with $u = \sqrt{1+x^3}$ and $\frac{\mathrm{d}v}{\mathrm{d}x} = 1$, so $v = x$ and $\frac{\mathrm{d}u}{\mathrm{d}x} = \dfrac{3x^2}{2\sqrt{1+x^3}}$:

$$\begin{aligned}\int 2\sqrt{1+x^3}\,\mathrm{d}x &= 2x\sqrt{1+x^3} - 2\int x\cdot\frac{3x^2}{2\sqrt{1+x^3}}\,\mathrm{d}x \\ &= 2x\sqrt{1+x^3} - \int \frac{3x^3}{\sqrt{1+x^3}}\,\mathrm{d}x.\end{aligned}$$

The last integral is exactly the second term of the integrand, so adding it on cancels it:

$$\int\left(2\sqrt{1+x^3} + \frac{3x^3}{\sqrt{1+x^3}}\right)\mathrm{d}x = 2x\sqrt{1+x^3} + c.$$

Differentiating $2x\sqrt{1+x^3}$ by the product rule gives back $2\sqrt{1+x^3} + \dfrac{3x^3}{\sqrt{1+x^3}}$, which confirms it.

Part (ii)

Split the integrand as $\dfrac{\sin x}{x} + \dfrac{2\sin x}{x^3}$. Neither piece has an elementary antiderivative on its own (the first is the famous Si integrand), so they must combine, just as in (i). Integrate the second piece by parts, with $u = \sin x$ and $\frac{\mathrm{d}v}{\mathrm{d}x} = 2x^{-3}$, so $v = -x^{-2}$:

$$\int \frac{2\sin x}{x^3}\,\mathrm{d}x = -\frac{\sin x}{x^2} + \int \frac{\cos x}{x^2}\,\mathrm{d}x.$$

Now integrate $\dfrac{\cos x}{x^2}$ by parts, with $u = \cos x$ and $v = -x^{-1}$:

$$\int \frac{\cos x}{x^2}\,\mathrm{d}x = -\frac{\cos x}{x} - \int \frac{\sin x}{x}\,\mathrm{d}x.$$

The $\int \frac{\sin x}{x}\,\mathrm{d}x$ cancels against the first piece, leaving

$$\int (x^2+2)\frac{\sin x}{x^3}\,\mathrm{d}x = -\frac{\sin x}{x^2} - \frac{\cos x}{x} + c = -\frac{\sin x + x\cos x}{x^2} + c.$$

As a check, differentiating $-\dfrac{\sin x}{x^2}$ gives $-\dfrac{\cos x}{x^2} + \dfrac{2\sin x}{x^3}$, and differentiating $-\dfrac{\cos x}{x}$ gives $\dfrac{\sin x}{x} + \dfrac{\cos x}{x^2}$; the $\cos x/x^2$ terms cancel and the sum is the integrand.

Part (iii)(a)

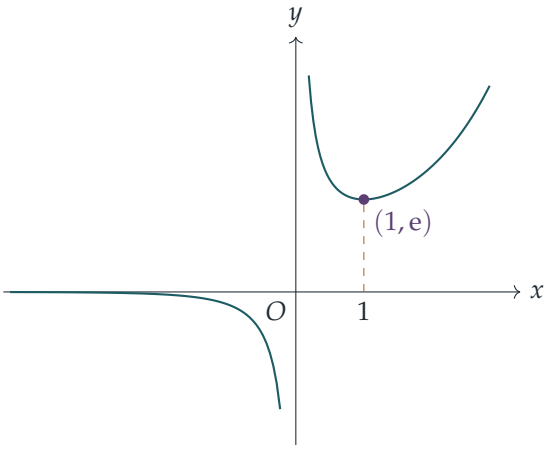
Let $y = \dfrac{\mathrm{e}^x}{x}$, defined for $x \neq 0$. By the quotient rule,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{x\mathrm{e}^x - \mathrm{e}^x}{x^2} = \frac{\mathrm{e}^x(x-1)}{x^2}.$$

This is zero only at $x = 1$, so the only stationary point is $(1, \mathrm{e})$. The derivative is negative for $x < 1$ ($x \neq 0$) and positive for $x > 1$, so $(1, \mathrm{e})$ is a minimum.

The rest of the shape:

- For $x < 0$, $y < 0$ and y is decreasing. As $x \to -\infty$, $y \to 0^-$ (the exponential wins), so the negative x-axis is an asymptote. As $x \to 0^-$, $y \to -\infty$.
- For $x > 0$, $y > 0$. As $x \to 0^+$, $y \to +\infty$; y falls to the minimum value e at $x = 1$, then increases to $+\infty$, faster than any polynomial.
- The y-axis is a vertical asymptote and the curve never meets either axis.



Part (iii)(b)

Integrate $\dfrac{e^x}{x^2}$ by parts, with $u = e^x$ and $v = -x^{-1}$:

$$\int \frac{e^x}{x^2}\,dx = -\frac{e^x}{x} + \int \frac{e^x}{x}\,dx, \qquad \text{so} \qquad \int \left(\frac{e^x}{x} - \frac{e^x}{x^2}\right) dx = \frac{e^x}{x} + c.$$

(Equivalently, $\frac{e^x}{x} - \frac{e^x}{x^2} = \frac{e^x(x-1)}{x^2}$ is the derivative we found in (a).)

? **Ask yourself** Is it legitimate to integrate between a and $2a$ at all?

Both integrands blow up at $x = 0$, and near 0 the integral of e^x/x^2 diverges, so the interval must avoid 0. This needs $a \neq 0$; then a and $2a$ have the same sign, so 0 is never between them, and the integrals are finite.

The two integrals are equal exactly when their difference is zero:

$$\left[\frac{e^x}{x}\right]_a^{2a} = \frac{e^{2a}}{2a} - \frac{e^a}{a} = \frac{e^a}{2a}\left(e^a - 2\right) = 0.$$

Since $e^a/(2a) \neq 0$, we need $e^a = 2$, so

$$a = \ln 2.$$

Part (iii)(c)

Exactly as in (b), for integers $m \neq n$ the two integrals can only be finite if 0 is not in the closed interval between m and n, so m and n are non-zero and of the same sign. Their difference is

$$\int_m^n \frac{e^x}{x}\,dx - \int_m^n \frac{e^x}{x^2}\,dx = \frac{e^n}{n} - \frac{e^m}{m},$$

so we need two distinct integers of the same sign at which $f(x) = e^x/x$ takes the same value. In other words, a horizontal line must cut the graph from (a) at two integer values of x.

Ask yourself The curve for $x > 0$ is not one-to-one, since it comes down and goes back up. So why can't two positive integers give the same height?

Because the turning point is at $x = 1$, which is itself the smallest positive integer. Taking the cases in turn:

- *Both negative.* From (a), f is strictly decreasing on $x < 0$, so $m \neq n$ gives $f(m) \neq f(n)$.
- *Both positive.* Then $m, n \geq 1$, and $f'(x) > 0$ for $x > 1$, so f is strictly increasing on $[1, \infty)$. Again $m \neq n$ gives $f(m) \neq f(n)$.

A horizontal line can meet the positive branch twice, but one of the two meeting points then lies in $0 < x < 1$, where there are no integers. So no pair of distinct integers works, and it is impossible, as required.

Checks

- For (i), at $x = 1$ the antiderivative is $2\sqrt{2}$, and a numerical value of the integral over $[0, 1]$ is $2.8284\ldots$, matching.
- For (ii), the integral over $[1, 10]$ evaluates numerically to $1.4711\ldots$, the same as the change in $-(\sin x + x \cos x)/x^2$.
- For (iii)(b), with $a = \ln 2$ both integrals come to about 1.9224. Note also that $f(\ln 2) = 2/\ln 2 = f(2 \ln 2)$: the two limits sit at equal heights on either side of the minimum, as the sketch suggests they must.
- For (iii)(c), the closest any two integers come is $f(1) = \mathrm{e} \approx 2.718$ against $f(2) = \frac{1}{2}\mathrm{e}^2 \approx 3.695$.

Common slips

- In (b) and (c), forgetting that an interval containing 0 makes the integrals meaningless, and so missing why m and n must have the same sign.
- In (c), arguing "f is one-to-one" without restriction: it isn't (for example $f(\ln 2) = f(2 \ln 2)$). The argument needs the turning point at $x = 1$.

Beyond STEP

Parts (i) and (ii) are cases where neither piece of the integrand has an elementary antiderivative, yet the sum does. Liouville's theorem in differential algebra makes precise which functions have elementary antiderivatives; e^x/x, $\sin x/x$ and $\sqrt{1 + x^3}$ do not, which is why the cancellation is essential. Part (iii)(c) is an injectivity argument, and analysis courses make such reasoning about monotonicity and the mean value theorem rigorous; see Abbott, *Understanding Analysis*.

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